

PALLET MERGE ENHANCEMENT

To-Be Functional Blueprint

Client-Specific Requirement Enhancement

PREPARED FOR



Vifel Ice Plant & Cold Storage, Inc.
Cold-Storage Warehouse Management System

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1.0

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Confidential

Prepared by **Elyon Solutions International Inc.**

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1.0	04 Aug 2026	Elyon Solutions	Initial scope release issued for Vifel review.

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1. Executive Summary

This blueprint defines the to-be scope of the Pallet Merge Enhancement, a client-specific extension to Vifel's warehouse management system. It allows designated clients to consolidate several products or batches onto a single physical pallet — a practice their operations already follow — while keeping Vifel's pallet count, occupancy, and billing figures accurate.

Today the system treats every received line as a separate pallet. For clients who legitimately place several items on one pallet, this over-states the number of physical pallets in storage and distorts occupancy-based billing. The enhancement introduces a controlled “merge” action, a matching “un-merge”, and the supporting configuration, identifiers, and reporting so that one physical pallet is always counted once — and only once.

AT A GLANCE

- Opt-in per client — every client is unaffected until merging is deliberately switched on.
- One physical pallet is counted once, no matter how many products sit on it.
- Fully reversible — any merge can be un-merged and the count self-corrects.
- Withdrawal, occupancy, and billing all read the true physical pallet count.
- No change whatsoever for clients who do not use merging.

2. Business Background & Rationale

As a cold-storage third-party logistics provider, Vifel measures storage and bills occupancy by the pallet. The standard system assumes one received line equals one new pallet. That assumption holds for most cargo, but not for clients who consolidate small lots — several products or production batches — onto a single pallet to use space efficiently.

Without a controlled way to record this, staff either over-count pallets (inflating occupancy and billing) or apply manual workarounds that are error-prone and leave the ledger inconsistent. The enhancement replaces those workarounds with a deliberate, auditable action that reflects what is physically on the floor.

- Accurate occupancy and billing — charge for the pallets that physically exist.
- A single, consistent method for staff, replacing ad-hoc manual corrections.
- A clear, reversible audit trail of what was merged, when, and by whom.

3. Scope of Enhancement

The enhancement is delivered as one installable capability covering seven grouped areas. Each area below states its intent and the key business requirements.

3.1 Client Configuration

Merging is governed entirely from the client record and is off by default; nothing changes for a client until it is enabled and configured.

- A master switch, “**Can Merge Pallets**”, turns the capability on for one client at a time.

- Two mutually-exclusive modes: a single Fixed Merge Pallet (one dedicated pallet reused every receipt), or Multiple Pallet Support (several special pallet types).
- An optional “Include Regular Pallets” setting also allows merging onto ordinary stock pallets.
- Optional per-client capture of Lot No. and Batch # (see 3.5).

3.2 Pallet Series Identifiers (PSI)

Every physical pallet carries exactly one unique series number. Merge clients use dedicated series types so special pallets are recognisable and never collide with ordinary numbering.

- One series equals one physical pallet — a system-wide integrity rule.
- Dedicated special series types are created automatically for a Multiple-mode client.
- Numbers are drawn and recycled automatically, never reused across the wrong pool.

3.3 Merge on Receiving

During receiving, a line can be merged onto a pallet that is already in stock, or onto a newly-created special pallet. Merging is available from both the Pallet Breakdown and the guided Magic Wizard, and behaves identically on each. The action is staged — nothing is committed until the user confirms.

Situation	Pallet count impact
Adding stock onto a pallet that already holds stock	+0 (no new pallet)
First stock onto an empty dedicated (Fixed) pallet	+1
Joining another line on the same receipt	+1 (the shared pallet) (New Pallet) +0 If already in Stocks
Starting a brand-new special pallet	+1

- The merged line adopts the target pallet’s series and location automatically.
- A note is recorded on the receiving report whenever the pallet count is intentionally not incremented.

3.4 Un-merge

Any merged line can be separated again. The pallet count self-corrects and no captured data is lost.

- Un-merge is offered wherever merge is — the Pallet Breakdown and the Magic Wizard.
- The separated line reverts to its own pallet series and location.
- The pallet count is restored as if the merge had not happened.

3.5 Lot No., Batch # & Prodcod Tracking

Optional client reference fields are captured at receiving and carried through stock, withdrawal, and reports, with a standardised product code frozen at validation.

- Lot No. and Batch # entry are shown only for clients configured to use them.
- A Prodcod is composed automatically (expiration date, batch, and building) and shown read-only on the withdrawal.
- These references are available as optional columns in the stock and withdrawal views.

3.6 Withdrawal & Pallet-Count Integrity

A merge pallet may be withdrawn from in part, because it deliberately carries several products. The pallet is only counted as leaving storage when it is physically emptied.

- Partial withdrawal is allowed — take one product and leave the rest — without forcing a return document.
- The pallet counts as one leaving (−1) only on the withdrawal that empties it.
- For multi-truck shipments, the pallet is counted against the truck that empties it, not the first truck to draw from it.

3.7 Reports

Every pallet-based figure reflects the merged reality — one physical pallet is counted once across all documents and reports.

- Receiving and Withdrawal printed pallet counts, occupancy, and billing each count a merged pallet once.
- Where enabled, the Lot No., Batch #, and Prodcodex appear as tracking columns on the relevant documents.

4. Out of Scope

The following are intentionally not part of this enhancement:

- Clients without merging enabled — no change to their receiving, withdrawal, or pallet counts.
- Standard (non-merge) pallet handling — unchanged.
- Blast-freeze operations — unaffected.

5. Assumptions & Dependencies

- Per-client configuration is maintained by authorised Vifel administrators.
- Each physical pallet is represented by a single package and series.
- The enhancement is installed once and remains installed.
- Supporting database automations are updated during deployment.
- Users receive brief orientation on the merge and un-merge actions.

6. Glossary

Term	Meaning
PSI (Pallet Series Identifier)	The unique number identifying one physical pallet.
Merge (+0)	Adding stock onto an existing pallet without increasing the pallet count.
Same-receipt join	Two lines on one receipt sharing a pallet; counted once.
Fixed Merge Pallet	A single dedicated pallet a client reuses on every receipt.
Multiple Pallet Support	Mode allowing several dedicated special pallet types.

Term	Meaning
Un-merge	Reversing a merge; the pallet count self-corrects.
Pallet Breakdown	The receiving screen listing each pallet line.
Magic Wizard	The guided receiving-encoding screen.
RR / WR	Receiving Report / Withdrawal Report.
Prodcode	Standardised product/batch code frozen at validation.

7. Acceptance & Sign-off

This document represents the agreed scope of the Pallet Merge Enhancement. The following approvals are required before the build is accepted as complete.

Role	Name / Organisation	Signature	Date
Prepared by	Elyon Solutions International Inc.		
Approved by	Vifel — Warehouse Manager		
Approved by	Vifel — Inventory Supervisor		